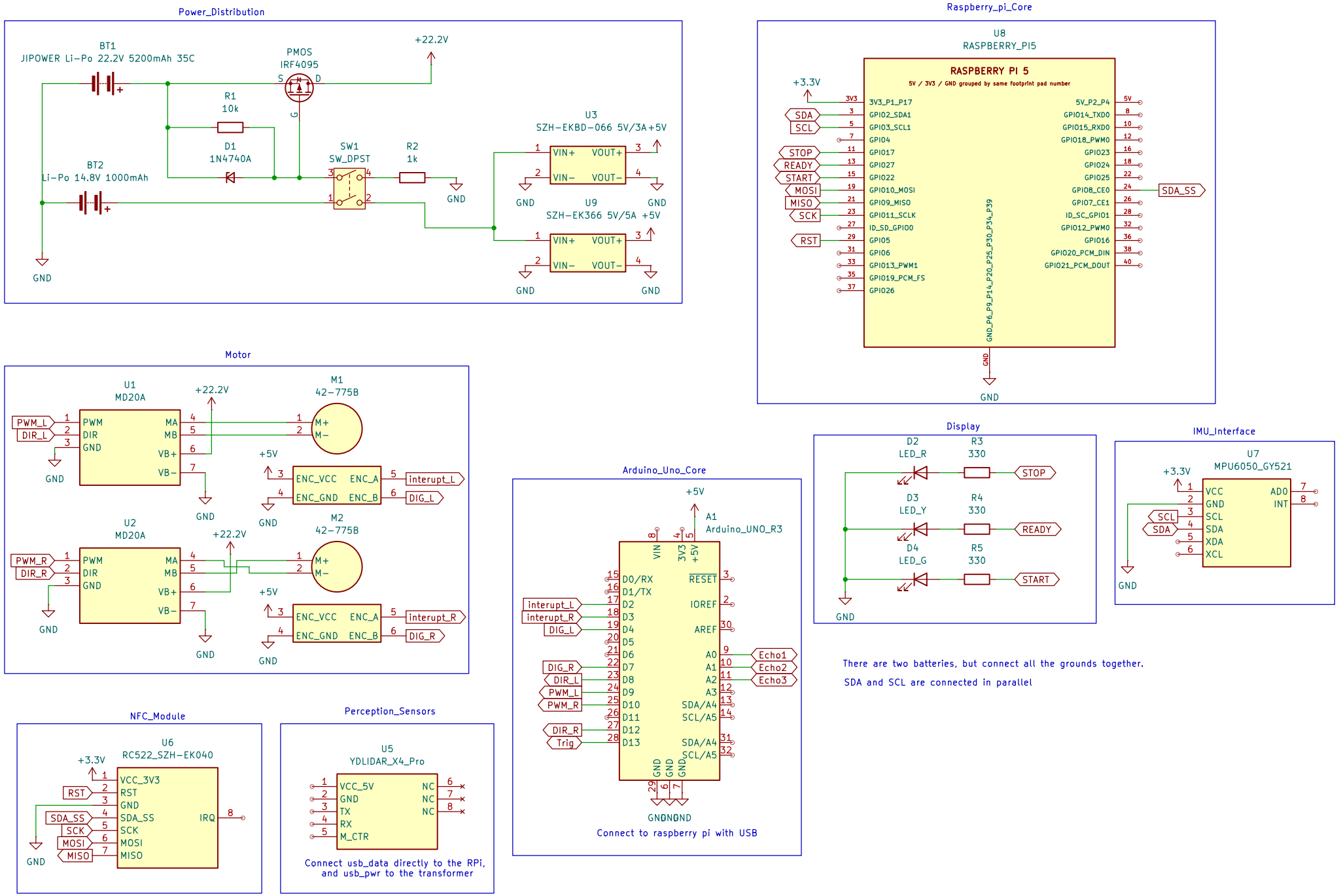
Power_Distribution
BT1
JIPOWER Li-Po 22.2V 5200mAh 35C
PMOS
IRF4095
+22.2V
R1
10k
D1
1N4740A
BT2
Li-Po 14.8V 1000mAh
SW1
SW_DPST
R2
1k
GND
U3
SZH-EKBD-066 5V/3A +5V
VIN+
VIN-
VOUT+
VOUT-
U9
SZH-EK366 5V/5A +5V
Raspberry_pi_Core
U8
RASPBERRY_PI5
RASPBERRY PI 5
5V / 3V3 / GND grouped by same footprint pad number
+3.3V
3V3_P1_P17
GPIO2_SDA1
GPIO3_SCL1
GPIO4
GPIO17
GPIO27
GPIO22
GPIO10_MOSI
GPIO9_MISO
GPIO11_SCLK
ID_SD_GPIO0
GPIO5
GPIO6
GPIO13_PWM1
GPIO19_PCM_FS
GPIO26
5V_P2_P4
GPIO14_TXD0
GPIO15_RXD0
GPIO18_PWM0
GPIO23
GPIO24
GPIO25
GPIO8_CE0
GPIO7_CE1
ID_SC_GPIO1
GPIO12_PWM0
GPIO16
GPIO20_PCM_DIN
GPIO21_PCM_DOUT
GND_P6_P9_P14_P20_P25_P30_P34_P39
SDA
SCL
STOP
READY
START
MOSI
MISO
SCK
RST
SDA_SS
Motor
U1
MD20A
M1
42-775B
PWM_L
DIR_L
PWM
DIR
MA
MB
VB+
VB-
M+
M-
+5V
ENC_VCC
ENC_GND
ENC_A
ENC_B
interupt_L
DIG_L
U2
MD20A
M2
42-775B
PWM_R
DIR_R
interupt_R
DIG_R
Display
D2
LED_R
R3
330
D3
LED_Y
R4
330
D4
LED_G
R5
330
IMU_Interface
U7
MPU6050_GY521
VCC
GND
SCL
SDA
XDA
XCL
ADO
INT
Arduino_Uno_Core
A1
Arduino_UNO_R3
VIN
3V3
+5V
D0/RX
D1/TX
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
RESET
IOREF
AREF
A0
A1
A2
A3
SDA/A4
SCL/A5
Echo1
Echo2
Echo3
Trig
GNDNDND
Connect to raspberry pi with USB
There are two batteries, but connect all the grounds together.
SDA and SCL are connected in parallel
NFC_Module
U6
RC522_SZH-EK040
VCC_3V3
RST
GND
SDA_SS
SCK
MOSI
MISO
IRQ
Perception_Sensors
U5
YDLIDAR_X4_Pro
VCC_5V
TX
RX
M_CTR
NC
Connect usb_data directly to the RPi,
and usb_pwr to the transformer